

Ko-Muping: Can you right-away compute the bias and dispersion, fast?

7 件のメッセージ

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月22日 5:58

To: Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Dear Ko and Muping, how long would it take you to compute the bias and variance? Can you do it today?

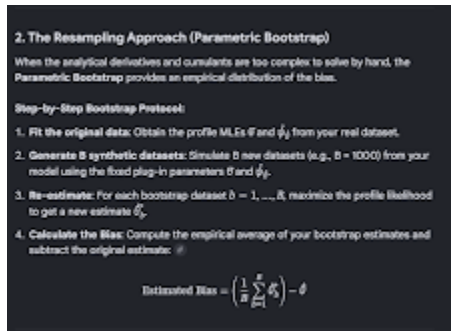
Computing the bias and dispersion means to generate many variations of the data preditec by our best-fit function (which is the data N_i we already have- namely variate them directly with a Gaussian distribution) , find the best fit from each of these new data sets and then find the average and dispersion of the resulting Φ_i at several energies.

I understand the box here in black - could not read Sec 3.4 of the paper sent by Muping (here too) because I do not understand right away the notation-

All nuisance parameters, including backgrounds h_i and b_i need to be kept constant too (as well as the best-fit for all $\delta\Phi_i$).

Best,
Graciela

添付ファイル 2 件



屏幕截图 2026-08-21 144510.png

98K



1-s2.0-S0047259X11002168-main.pdf

402K

Yasuda Koichiro <k.yasuda5160@gmail.com>

2026年8月23日 15:13

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Dear all

I talked with Muping just now. Here I show the bias in the sense of what Muping has sent.

Best,
Ko

2026年8月22日(土) 5:58 Graciela Gelmini <gelmini@physics.ucla.edu>:

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bias_1eV_3kgyear_ExpBkg.pdf
33K

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月24日 6:27

To: Yasuda Koichiro <k.yasuda5160@gmail.com>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Dear Ko, and Muping, THANKS!!!! Can you please show the same plot also as **percentage** of the "true value".

Assessing Smallness: Bias and dispersion (standard deviation) are typically evaluated relative to the true value locally in each bin as a fractional or Relative Bias_i = Bias_i / True value_i

Relative Uncertainty_i = sigma_i / True value_i.

Bias is small if Relative Bias_i <<1 . Dispersion is small if Relative Uncertainty_i <<1.

Bias is considered acceptable if Relative Bias_i² < \variance_j

We should also know the error these calculations have: Error on Bias_j = \sqrt{\variance_j}/

\sqrt{N} (N=number of toy models used) and

Error on Variance= variance_j \times \sqrt{2/(N-1)}

Definitions of bias and variance?:

I did not understand what scenarios A and B are. But what you plotted now seems to be correct (notes I sent yesterday - attached here again)

bias (E_j)= (average of Phi_j the toy models results at E) - ("true"value(E) = or us the value of the initial best fit.)

What is the definition of the dispersion sigma you used? Is

sigma (E_j)²= variance (E) = average of (value of each toy model (E)- average of all toy models (E))²

where for the average of Phi_j one uses N and for the average in the variance one uses N-1 instead of N in the denominator- (although for large N this does not make a difference.

Best,
Graciela

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MonteCarlo bias, variance ,Total Mean Squared Error (MSE) of an unfolding method.pdf
900K

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月24日 6:36

To: Yasuda Koichiro <k.yasuda5160@gmail.com>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

What we need is the bias in percentage and the dispersion in percentage in the same figure at each energy, not of the bias plus-minus the dispersion at each energy. As I mention below Bias is small if Relative Bias_i <<1 .

Dispersion is small if $\text{Relative Uncertainty}_i \ll 1$.

Bias is considered acceptable if $\text{Relative Bias}_i^2 < \text{variance}_i$

Graciela

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Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月24日 6:45

To: Yasuda Koichiro <k.yasuda5160@gmail.com>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Maybe it would be good to plot separately, in a different plot, the ratio $\text{Bias}^2 / \text{variance}$ (both are positive numbers) if this ratio is small the bias is considered small. Judging by eye I think we have a large bias. But it is difficult to see at higher energies in the plot Ko sent.

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Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月24日 8:28

To: Yasuda Koichiro <k.yasuda5160@gmail.com>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Besides plotting in a figure the relative bias and relative dispersion as I said before, I think it may also be good to plot in a different figure their squares since it is the ratio of the squares that determines if the bias is "small" (will see later what is better to show in the paper).

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Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月24日 8:58

To: Yasuda Koichiro <k.yasuda5160@gmail.com>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

In optimized Tikhonov-regularized unfolding (where the MSE is minimized) it seems that ratio ($\text{bias}^2 / \text{variance}$) is about 1. See the notes attached. Best, Graciela

PS Will go now to a hike and will return at about 21 to 22 hours in LA. So I will not be in contact until then

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In optimized Tikhonov-reg-ratio of bias-square and varian is about 1.pdf

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